

Investigation of the surface treatment of GaAsSb/Si using current- and capacitance-voltage measurements

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ABSTRACT

In this study, the effects of surface treatments on the bonded interface between novel III-V absorber material GaAs_{0.51}Sb_{0.49} latticed-matched to InP and Si are experimentally investigated. Both hydrophobic- and hydrophilic-based surface treatments are performed on target Si substrates prior to transfer-printing-based bonding of GaAsSb. Both current-voltage and capacitance-voltage measurements on these devices suggest that interface oxides, which contribute to robust bonding between GaAsSb and Si, block photoinduced charge transport. Unlike hydrophilic O₂ plasma treatment, hydrophobic HF wet etching on Si produces an optimized interface with less defects and facilitates charge transport. After the HF-bonded sample was treated under vacuum right before transfer, the resulting *p-i-n* photodiodes based on *p*-GaAsSb/*i*-GaAsSb/*n*-Si exhibits device performance comparable to monolithic GaAsSb diodes directly grown on InP substrates. Under 1550 nm illumination, the printed diode outputs a V_{OC} of 230 mV, while the monolithic GaAsSb diode outputs a V_{OC} of 320 mV.

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INTRODUCTION

Si-based photonic integrated circuits (PICs) are gaining tremendous attention due to their convenient implementation with Si CMOS manufacturing.¹ However, Si does not deliver satisfying optoelectronic (such as detection) functionality at short-wave infrared wavelengths due to its bandgap.² The heterogeneous integration of III-V semiconductors on Si provides a solution to this fundamental problem.^{3–5} Due to significant lattice and thermal mismatches between III-V semiconductors and Si, the direct growth of the former on the latter using modern growth technologies induces abundant cracks and defects that are detrimental to film quality.^{6,7}

Wafer bonding and transfer printing techniques are frequently used in heterogeneous integration, and these techniques can improve film quality by bonding the as-grown III-V semiconductor with Si under rapid thermal annealing (RTA).^{8,9} In contrast

to wafer bonding, transfer printing involves a polydimethylsiloxane (PDMS) stamp that can pick up a III-V semiconductor film with thickness of a few hundred nanometers from a source substrate and print it on a target Si substrate. Because of strong van der Waals forces acting between the very thin III-V semiconductor film and Si, bonding can be achieved without surface activation.^{10,11} Here we use the transfer printing technique to achieve III-V integration on Si considering its ability to integrate a single device into PICs.

In regard to III-V semiconductors, a novel absorber GaAsSb latticed-matched to InP has recently gained serious attention due to its efficacy in delivering a commendable absorption ability of 1550 nm photons, which is frequently used in telecommunications and sensors.¹² For 1550 nm wavelength detection, separate absorption, charge, and multiplication (SACM) avalanche photodiodes (APDs) have independent photon absorption and impact ionization multiplication processes, which can be optimized individually

to improve both noise and speed performance.¹³ Even without the ability to detect 1550 nm wavelength, the very low excess noise factor of Si itself provides high gain and low noise in charge multiplication, making it a promising multiplier in SACM APDs.¹⁴ Compared to InGaAs, the conduction band offset of GaAsSb with Si is much smaller, allowing for faster electron transport.¹⁵ Combining all the factors above, the GaAsSb/Si heterostructure has significant potential for application in SACM APDs.^{16,17}

In 1997, Hawkins *et al.* demonstrated an infrared InGaAs/Si APD, achieving a gain-bandwidth product exceeding 300 GHz.¹⁸ The *n*-InGaAs layer was fused to the Si surface by wafer bonding at temperatures of 650 °C for 10 min in a H₂ environment. A low stress and high yield wafer bonding technology was then successfully employed in the fabrication of reliable *p-i-n* photodiodes and APDs. The fabricated InGaAs/Si photodiodes delivered dark current and responsivity that were comparable to commercially available InP-based *p-i-n* photodiodes and APDs.¹⁹ Nevertheless, this bonding technique involved high-temperature hydrophobic wafer bonding at above 500 °C, which would degrade the performance of other components in PICs. In 2016, Um *et al.* utilized the transfer printing technique to bond InGaAs nanomembranes to Si substrates via van der Waals forces. Transfer printed InGaAs/Si photodiodes exhibited a high rectification ratio (7.73×10^4 at ± 3 V) and low leakage current (7.4×10^{-5} A/cm² at -3 V) in a dark state.²⁰ However, the InGaAs/Si photodiode was not particularly responsive to 1550 nm wavelength. In 2024,

Muduli *et al.* demonstrated a GaAsSb/Si photodiode via transfer printing.¹⁶ The GaAsSb/Si photodiodes delivered a substantial 1550 nm wavelength photocurrent at room temperature. However, the influence of oxides, voids, defects, and surface treatments on the bonded interface quality, and in turn the performance of photodiodes, has not been studied comprehensively in transfer printing.

In this experimental work, monolithic GaAsSb/InAlAs/InP (sample A) and heterogeneously integrated GaAsSb/Si (sample B) *p-i-n* photodiodes were investigated. For the latter, HF- or O₂ plasma treatment was applied on Si prior to printing to estimate qualitatively the influence of interface oxide on charge carrier transport. Comprehensive characterization of the two structures was performed and compared to identify the influence of the interface on charge transport.

EXPERIMENTAL METHODS

All III-V structures were grown by molecular beam epitaxy (MBE). This study utilized one MBE-grown structure (sample A) and one transfer printed structure (sample B) as schematically shown in Fig. 1(a). On top of InP substrates, 350 nm *n*⁺-InAlAs was deposited as a sacrificial layer for transfer printing. Unintentionally doped (UID) lattice-matched GaAsSb with a thickness of 1800 nm was then grown as an absorber. The final *p*⁺-GaAsSb was grown as a contact layer for ohmic contact formation.

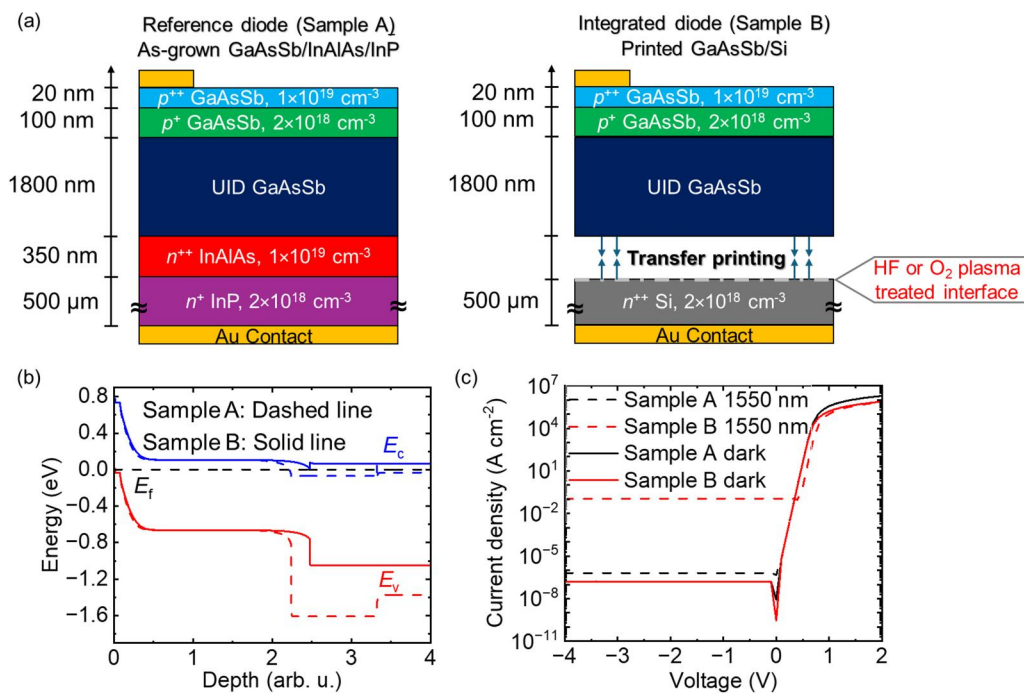


FIG. 1. (a) Schematic representation of device structure for as the grown GaAsSb/InAlAs/InP diode (sample A) and GaAsSb/Si diode (sample B); (b) band diagram simulation of both diodes under equilibrium; and (c) simulated dark current and photocurrent of both diodes under 1550 nm wavelength illumination.

Factors such as oxide thickness and voids cause inconsistencies in diode performance. Because transfer printing was performed manually at ambient humidity and temperature greatly influence the bonding interface. Hence, diodes using these materials were fabricated under the same conditions for a fair comparison. The surface of target Si substrates was prepared with both HF- or O₂ plasma. GaAsSb from the source substrate was then printed on Si. GaAsSb/Si *p-i-n* diode fabrication was ended by RTA treatment at 300 °C for 5 min.

GaAsSb mesas were defined by standard photolithography, and a subsequent BCl₃/Ar dry etch. Details of the transfer printing process can be found in a previous report.¹⁶ Before transfer printing, the target Si substrate was either immersed in HF (49%) for 5 min or treated with an O₂ plasma in a flow of 20 SCCM and a power of 45 W for 5 min. After printing on Si, the GaAsSb film contained tethers used to suspend GaAsSb during transfer printing. Because transfer printing was performed manually, the printed devices deviated slightly from their original location. Thereafter, photolithography and dry etching were performed to remove the tethers and create a new cylindrical GaAsSb mesa for the final metal deposition.

RESULTS AND DISCUSSION

The band diagrams under equilibrium of the two devices are displayed in Fig. 1(b). The simulated UID GaAsSb is set to be *n*-type with a doping of $5 \times 10^{15} \text{ cm}^{-3}$. The depletion region is centered on the UID GaAsSb in both devices. Sample B delivers smoother conduction and valence band transition compared to sample A. The simulated dark current and photocurrent of the two devices under 1550 nm illumination are displayed in Fig. 1(c). Both the samples deliver the same dark current, which is explained by the fact that they have the same depletion region. The slight

difference of dark current under forward bias may result from the difference in series resistance between InP and Si. Interestingly, sample B delivers a much higher photocurrent than sample A. This can be further explained by the slower charge collection due to the hole barrier that InAlAs creates between GaAsSb and InP in sample A. This was confirmed by the photocurrent simulation results in Fig. S1 in the [supplementary material](#), where photocurrents are compared under different hole barriers.

Figure 2(a) shows the optical microscope image of a fully processed diode from sample B. The current-voltage (*I*-*V*) characteristics of sample A and sample B are displayed in Fig. 2(b), all the diodes are shown to deliver an obvious rectifying current as expected. Sample A yields an ideality factor of 1.5 irrespective of current-voltage (*I*-*V*) sweeps. In the first sweep, HF-treated sample-B exhibits an ideality factor of 1.2 and O₂-plasma treated one provides 1.8. This indicates that O₂ plasma treatment induces serious trap-assisted nonradiative recombination in the diode, which is further evidenced by the difference in dark current under reverse bias.

Consecutive voltage sweeps are performed on these three diodes and time gap between these two measurements is around 3 s. Sample A devices deliver consistent *I*-*V* curves in these two sweeps. It is observed that sample B diode currents during their second sweep are larger than that of the current measured during the first forward sweep at a given small voltage. With an increase in voltage, the linear region of the logarithmic *I*-*V* curve disappears, and the current finally reaches the high-injection region. In the second sweep, HF-treated sample B exhibits an ideality factor of 5.2 and O₂-plasma treated one provides 6.3, which both exceed the theoretical limit. We speculate that interface oxides contribute to the inconsistent *I*-*V*s in dark conditions, considering the fact that the current inconsistency becomes more pronounced after O₂ plasma treatment on Si. This forward current inconsistency could

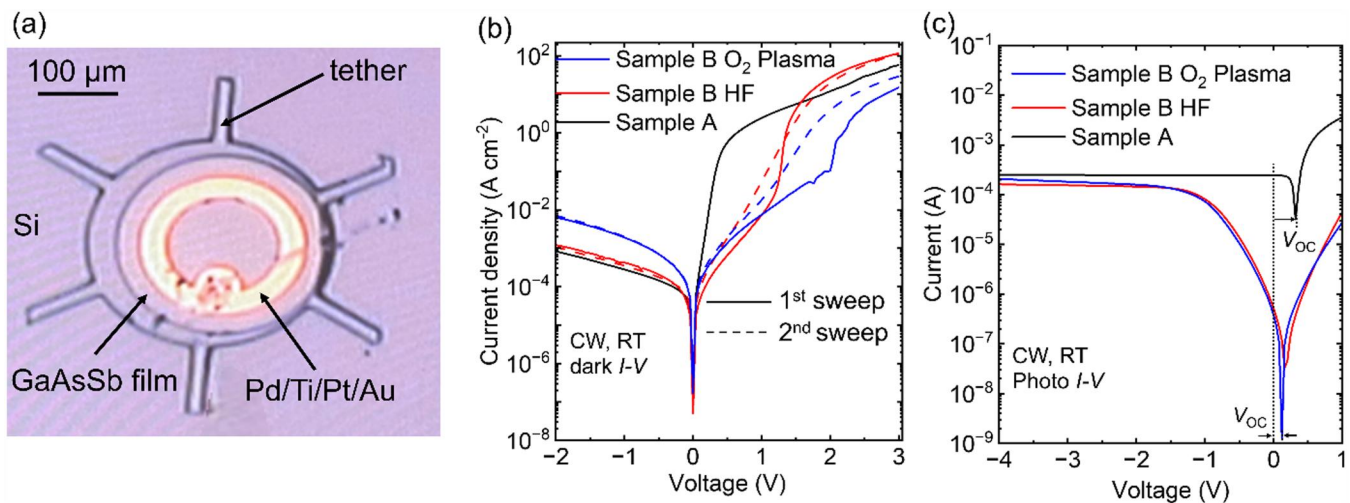


FIG. 2. (a) Optical microscope image of the fully processed circular devices based on sample B with a diameter of $260 \mu\text{m}$, (b) current of diodes under dark conditions, and (c) photocurrent of diodes under 1550 nm wavelength illumination. Sample A and sample B are of different sizes and illuminated with the same optical power. We believe it would be more reasonable to compare their photocurrent instead of current density.

be possibly due to the carrier blocking effect induced by the thin interface oxide layer. During the first forward sweep, carriers are blocked by the oxide layer and eventually delay the turn-on of the diodes. In the second forward sweep, the carrier can be collected sooner, giving rise to a higher current than the first sweep. Additionally, dark I - V of sample B switches to its first sweep after diodes are relaxed. It indicates that charges in the interface dissipate over time. Under reverse bias, carrier densities are very low, and the influence on interface charge transport is negligible. Details of the diodes' I - V performance are summarized in Table I.

The currents through these three diodes under 1550 nm wavelength laser illumination with a power of 4.4 W are displayed in Fig. 2(c). All three diodes delivered prominent photocurrent as evident from comparing their currents under dark and light illumination. Sample A delivers flat current under reverse bias with an open circuit voltage (V_{OC}) of 320 mV,

$$V_{OC} = \frac{k_B T}{q} \ln \left(\frac{J_{ph}}{J_{ss}} + 1 \right),$$

where V_{OC} is open circuit voltage, k_B is the Boltzmann constant, T is temperature, q is charge, J_{ph} is photocurrent density, and J_{ss} is saturation current density. However, we see a drastic decrease in current for sample B when reverse voltage decreases to 0 V, which is unusual for common photodiodes. We attribute this to charge blocking from the interface oxide. The charge-blocking effect is discussed further in the light capacitance-voltage (C - V) measurement analysis. The HF-treated sample B delivers a V_{OC} of 160 mV, and the O_2 -plasma-treated sample B diode delivers a V_{OC} of 120 mV. This phenomenon of decreasing V_{OC} is consistently observed across different samples and devices. After enhancing the O_2 plasma power, the treated diode even delivers no V_{OC} as shown in Fig. S2 in the supplementary material.

To characterize the photoresponsivity of the devices, external quantum efficiency (EQE) measurements were performed under controlled optical and electrical bias conditions. A 1550 nm continuous-wave laser source was used to generate the incident optical signal. The laser output was coupled into an optical fiber and delivered to the device as a 23.6 μ m-diameter circular beam that underfilled the active area, ensuring the entire optical spot remained fully confined within the photoactive region. The laser beam was modulated, and a 90/10 fiber splitter diverted a small fraction of the optical power to a Thorlabs PM100D power meter

for real-time monitoring, while the remaining signal illuminated the device under test. Electrical biasing and photocurrent measurements were performed using a Keithley 2450 Source Measure Unit (SMU). The external quantum efficiency was calculated according to

$$EQE = \frac{I_{ph} * h\nu}{q * P_{opt}},$$

where I_{ph} is the measured photocurrent, h is the Planck constant, ν is the optical frequency corresponding to 1550 nm, q is the elementary charge, and P_{opt} is the measured incident optical power. In Fig. 3(a), sample A exhibits an EQE of 21.5%, which is constant irrespective of both reverse bias and incident optical power, indicating efficient carrier collection across a broad range of optical injection levels. In contrast, sample B, fabricated by transferring thin GaAsSb films onto Si substrates, exhibits a reduction in EQE with increasing optical power from 0.26 to 0.41 mW as shown in Figs. 3(b) and 3(c). This reduction in EQE at higher optical powers can be attributed to trap-assisted recombination processes becoming more significant at elevated carrier densities due to interface defects and bonding-induced trap states introduced during transfer printing. For sample B, a significant increase in EQE with applied reverse bias is observed, consistent with the I - V characteristics in Fig. 2(b). Also, their EQE difference is around 1%, which is tolerable and can be ascribed to the subtle difference in material quality and fabrication conditions.

Capacitance-frequency (C - f) measurements, as shown in Fig. S3 in the supplementary material, were performed on the three diodes to characterize their capacitance response to different sweeping frequencies under certain applied bias. Based on the C - f results, we choose a frequency of 1 MHz for carrying out C - V measurements on the diodes. The C - V results of the three diodes in the dark condition are displayed in Fig. 4(a). Their areas are normalized as a circle with a radius of 200 μ m. All the diodes deliver similar trends and closely match each other. This indicates that sample B interface does not introduce any significant capacitance compared to the junction capacitance. Also, the sample B interface does not alter the built-in electric field significantly, meaning the depletion region is still occupied by UID GaAsSb. The depletion region depth vs applied voltage for the three diodes are displayed in Fig. 4(b). All three diodes deliver a depletion region with a thickness of around 750 nm. After the reverse voltage increases to -4 V, the depletion region depth saturates at 1800 nm, which is consistent with the UID thickness. The background doping of UID GaAsSb in the three diodes are determined by extrapolation of CV data as shown in Fig. 4(c). All three diodes are then confirmed to have a similar doping of 1×10^{15} - 1.5×10^{15} cm $^{-3}$ within the UID GaAsSb, suggesting good material growth. When the depletion region saturates, the doping in sample A drastically increases to around 1×10^{19} cm $^{-3}$, which is consistent with the doping concentration of InAlAs. The doping in sample B drastically increases to around 2×10^{18} cm $^{-3}$, which is consistent with the doping concentration of Si.

Light C - V measurements were conducted to understand charge blocking phenomena at the interface. The light C - V

TABLE I. GaAsSb/InAlAs/InP and GaAsSb/Si diode properties extracted from I - V characteristics.

Device	Ideality factor	I_{on}/I_{off} ratio at ± 0.8 V	Series resistance (Ω cm $^{-2}$)
GaAsSb/InAlAs/InP	1.5	7200	0.24
GaAsSb/Si HF	1.2	46	0.02
GaAsSb/Si O_2	1.8	5.5	0.07

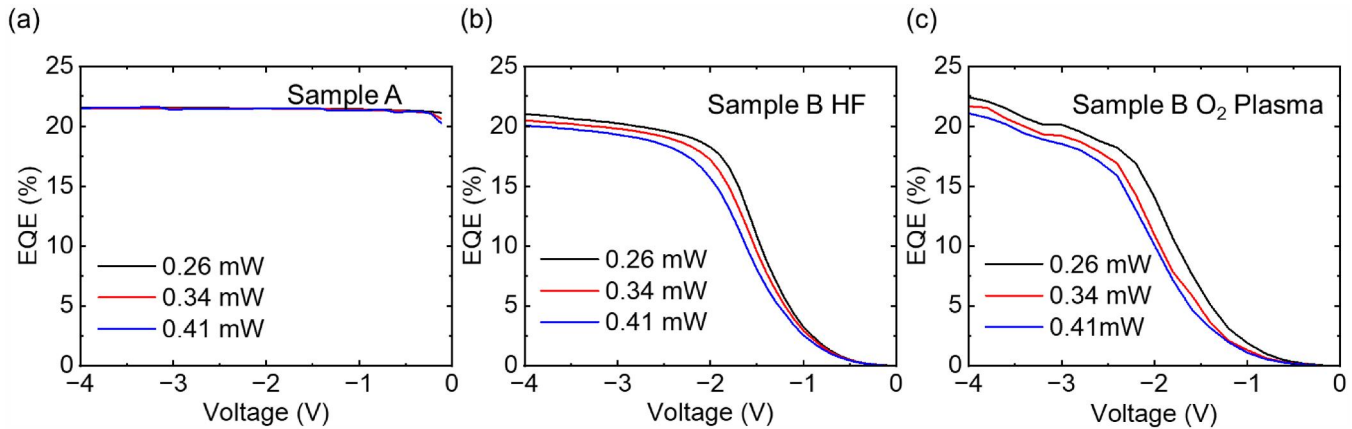


FIG. 3. EQEs of (a) sample A, (b) sample B with HF treatment, and (c) sample B with O₂ plasma treatment.

results of the three diodes are displayed in Fig. 5(a). Because sample A has a different size than sample B during fabrication, they deliver different capacitances in the graph. Sample A, at zero bias, delivers a capacitance of 24 pF in 1550 nm illumination, which is slightly higher than that of 21 pF in the dark. When reverse bias is applied, the capacitance in light condition instantly drops and converges with the capacitance in dark conditions. However, for sample B, both treatments deliver significantly increased capacitance under illumination. The HF treated diode delivers a capacitance of 100 pF and the O₂ plasma treated diode delivers a capacitance of 177 pF, compared to the capacitance of 9 pF without illumination. After the reverse bias was increased to -1 V, their capacitances drop drastically and gradually return to the value in dark conditions after the reverse bias voltage reaches -2 V. Interestingly, we observe a similar trend in the I - V of sample B in Fig. 2(b),

where the photocurrent is negligible at 0 V and then the current increases drastically at -1 V and saturates after the reverse voltage reaches -2 V.

We speculate that the high capacitance of sample B at 0 V under 1550 nm illumination stems from the light-generated carriers, which should be collected by the electrode but are blocked by the interface. Under reverse bias, the increased built-in electric field enables the carriers to tunnel through the interface and produce photocurrent. It is also found that the HF-treated diode delivers a smaller capacitance than the O₂-plasma-treated diode. This can be ascribed to the optimized interface in the HF-treated sample B. Here, we speculate that HF treatment modifies the interfacial oxide by reducing its thickness or decreasing its bandgap due to the removal of native SiO₂ on the Si surface prior to transfer printing. To simplify the band diagram simulation of the real device, SiO₂ was selected to represent the interface oxide.

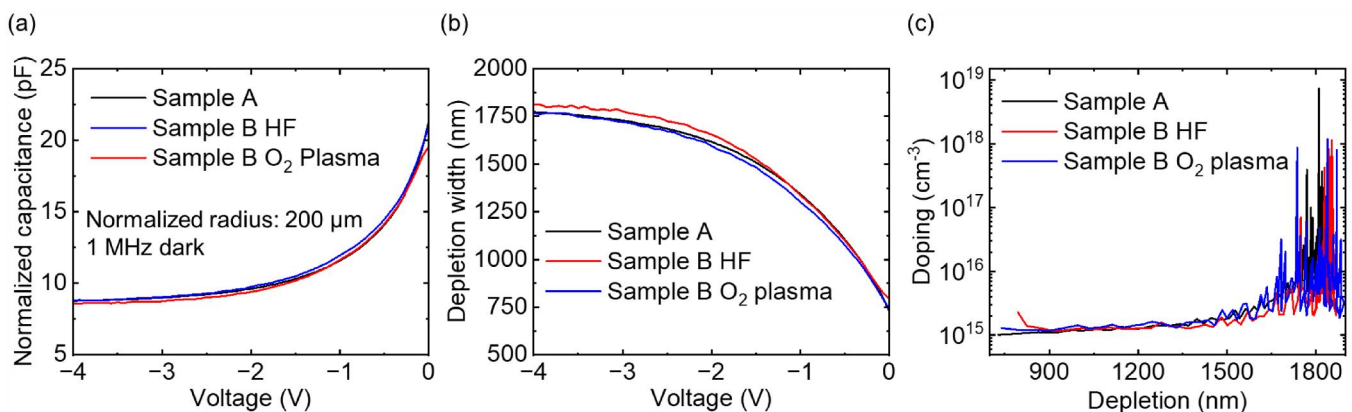


FIG. 4. (a) C - V characterization of the three diodes; (b) depletion width vs applied voltage for the three diodes; (c) doping vs depletion width for the three diodes.

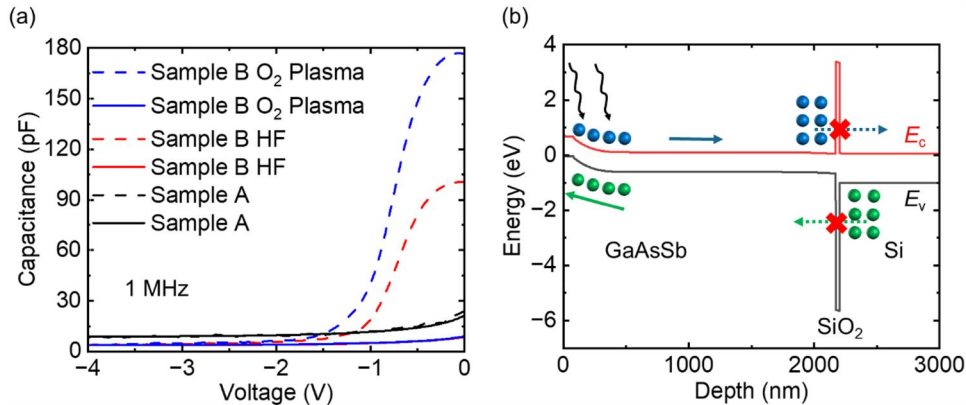


FIG. 5. (a) Dark (solid line) and light (dashed line) C - V characteristics for the three diodes; (b) GaAsSb/SiO₂/Si band diagram at zero bias.

Figure 5(b) displayed the GaAsSb/SiO₂/Si band diagram at zero bias. Without illumination, the small number of electrons driven by the built-in electric field can tunnel through the interface oxide, producing dark current. After illumination, abundant charge carriers are generated in the UID GaAsSb. Due to the large barrier formed by SiO₂, crowded electrons cannot tunnel through easily, forming a two-dimensional electron gas. This increases the charge density in the interface, which, in turn, increases the measured capacitance of the diodes. After reverse bias is applied, the increased built-in electric field endows the electrons with enough velocity to tunnel through the interface, forming the photocurrent.

In order to confirm this hypothesis, we optimized the fabrication process and put the PDMS/GaAsSb/Si sample in a vacuum chamber at a pressure of 75 mT for 3 min after PDMS/GaAsSb was printed on Si to enhance bonding and remove humidity, which can suppress the oxide formation. The HF-treated sample B delivers the same flat photocurrent as sample A does. The device exhibits photocurrent one order of magnitude lower than the device without vacuum treatment, which may come from the material quality difference. The I - V performance of the device is displayed in Fig. 6(a). The device produces a dark current density of approximately

$1 \times 10^{-3} \text{ A cm}^{-2}$ at -2 V , which is similar to sample A. Under 1550 nm illumination with a power of 4.4 mW, the photocurrent does not decrease drastically at 0 V and outputs a V_{OC} of 230 mV. The dark and light C - V s of the HF-treated sample B are compared in Fig. 6(b). No significant difference in capacitance is observed under these two conditions, suggesting that the charge-blocking effect of interface oxide disappears. Therefore, it can be confirmed that the oxide in the bonded interface causes a higher series resistance and can block charge transport under illumination.

In conclusion, we compared the photodiode performance of GaAsSb, composed of a p -type cladding layer and a UID absorption layer, on n -InP and n -Si substrates. Sacrificial-layer-wet-etch-based transfer printing was performed to fabricate a GaAsSb/Si diode, and the oxide in the interface was modified by treating Si with HF or O₂ plasma before transfer printing. Dark and photo I - V characterizations show that GaAsSb/Si diode delivers higher dark current and drastically decreased photocurrent near 0 V, which is different than simulated results. In light C - V measurements, the extremely high capacitance of the GaAsSb/Si diode under illumination at 0 V indicates that photogenerated carriers are trapped and cannot be collected, which can explain the drastically decreased

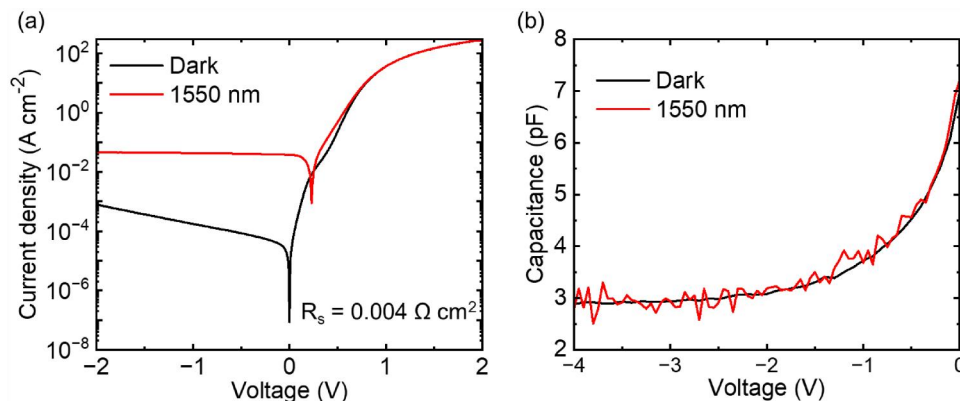


FIG. 6. (a) I - V characteristics and (b) C - V characteristics of HF-treated GaAsSb/Si diode treated under vacuum during transfer.

photocurrent. Interestingly, the HF-treated device delivers smaller capacitance than the O₂-plasma-treated device. We conclude that interface oxides may block photogenerated carriers at zero bias, and the trapped carriers can be collected by the application of a reverse bias. The combination of HF treatment on Si and vacuum condition during transfer printing can ameliorate this effect. Delicate interface characterizations are required to target the specific oxide that caused the charge-blocking effect. Further precise control of other factors such as temperature, humidity, and pressure during transfer printing and RTA conditions during annealing is required to fabricate devices with consistent performance.

SUPPLEMENTARY MATERIAL

The [supplementary material](#) includes the simulated dark- and photo *I*-*V*s of GaAsSb/InP, GaAsSb/InAlAs/InP, and GaAsSb/Si photodiodes; dark and photo *I*-*V*s of GaAsSb/Si diode treated with O₂ plasma at high RF power; and *C*-*f* measurements of GaAsSb/InAlAs/InP, GaAsSb/Si HF-treated, and GaAsSb/Si O₂-plasma treated diodes.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yongkang Xia and Manisha Muduli contributed equally to this paper.

Yongkang Xia: Conceptualization (equal); Formal analysis (lead); Investigation (equal); Writing – original draft (lead). **Manisha Muduli:** Conceptualization (equal); Formal analysis (equal); Investigation (equal); Writing – review & editing (equal). **Seunghyun Lee:** Formal analysis (supporting); Methodology (supporting); Writing – review & editing (supporting). **Nathan Gajowski:** Investigation (supporting); Writing – review & editing (supporting). **Sophie Mills:** Formal analysis (supporting); Investigation (supporting); Writing – review & editing (supporting). **Naga Swetha Nallamothu:** Investigation (supporting); Writing – review & editing (supporting). **Sk Shafaat Saud Nikor:** Formal analysis (supporting); Investigation (supporting); Writing – review & editing (supporting). **Rachel L. Adams:** Investigation (supporting); Writing – review & editing (supporting). **Ronald M. Reano:** Funding acquisition (supporting); Supervision (supporting); Writing – review &

editing (supporting). **Steven A. Ringel:** Formal analysis (supporting); Funding acquisition (equal); Project administration (lead); Resources (equal). **Sanjay Krishna:** Conceptualization (equal); Formal analysis (supporting); Funding acquisition (supporting); Supervision (supporting); Validation (supporting); Writing – review & editing (supporting). **Shamsul Arafin:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (lead); Resources (lead); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The supporting data for the findings of this study are available from the corresponding author upon reasonable request.

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